

Project Description: Multigraded Homological Algebra and Geometry

This proposal involves research in commutative algebra and algebraic geometry with several connections to computation and combinatorics, as well as a wide array of broader impacts.

- § 1 Results of Prior NSF Support.
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1. Results of Prior NSF Support

From 2020 to 2022, the PI held an NSF Postdoctoral Research Fellowship (MSPRF DMS-2002239, \$150,000), *Asymptotic Syzygies in Algebraic Geometry*. Since research projects and journal timelines extend well past the award period, the numbers below span 2020–2025. During this time, the PI substantially advanced homological algebra on toric varieties (Section 1.1) and the cohomology of moduli spaces (Section 1.3), posting 7 preprints to the arXiv [31, 33, 32, 21, 20, 5, 34] and publishing 7 articles, including in *Geometry & Topology* [20], *Journal of Algebra* [31], and *Experimental Mathematics* [35]. The sections below detail these results and their intellectual merit.

1.1 Eisenbud-Goto on Products of Projective Space The Castelnuovo–Mumford regularity of a projective variety $X \subset \mathbb{P}^r$ is a measure of the complexity of X given in terms of the vanishing of certain cohomology groups of X . Its centrality to commutative algebra rests on the fact that a single integer controls cohomology vanishing, the degrees of generators and syzygies, and the eventual linearity of resolutions.

Theorem 1.1. [55] *Let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^n$ and $M = \bigoplus_{e \in \mathbb{Z}} H^0(\mathbb{P}^n, \mathcal{F}(e))$ the corresponding section module. The following are equivalent:*

- (1) M is d -regular in the sense of [92];
- (2) $\beta_{p,q}(M) = 0$ for all $p \geq 0$ and $q > d + i$; and
- (3) $M_{\geq d}$ has a linear resolution in the sense of Eisenbud-Goto.

Nearly every use of regularity in the literature – regularity of powers of ideals [48, 81], the Eisenbud–Goto conjecture [MP18], the construction and boundedness of Hilbert schemes [67, 65], Gröbner basis complexity [8], etc. – passes through this equivalence in a crucial way.

A toric variety X is a well-behaved compactification of a torus and, under mild hypotheses, can be written as a quotient $X = (\mathbb{C}^n \setminus \mathbb{V}(B))/(\mathbb{C}^\times)^r$, where B is the *irrelevant ideal* of X ; for $X = \mathbb{P}^n$, this is $B = \langle x_0, \dots, x_n \rangle$. Cox’s theorem gives a correspondence between coherent sheaves on X and B -saturated $\mathbb{Z}^r$ -graded modules over the $\mathbb{Z}^r$ -graded polynomial ring $S = \mathbb{C}[x_1, \dots, x_n]$, known as the Cox ring of X [46]. This yields a rich dictionary translating questions about the geometry of X into multigraded commutative algebra. However, this dictionary is far subtler than in the standard graded case, in ways that have driven much recent activity: virtual resolutions and resolutions of the diagonal [11, 72, 26, 60], generalizations of the BGG correspondence [54, 28], and multigraded local cohomology [18, 42], to name a few.

Extending Castelnuovo–Mumford regularity beyond projective space has long been an active area of research. Early developments include Benson’s treatment of nonstandard $\mathbb{Z}$ -graded rings [9], later developed by Symonds in his groundbreaking work on invariant theory [100, 101], and Hoffman and Wang’s theory for products of two projective spaces [73]. In the toric setting, Maclagan and Smith gave a cohomological definition of multigraded regularity [87] that subsumes the weighted

and biprojective theories and has become the standard framework for extending the classical theory to toric varieties.

Throughout, X denotes a smooth projective toric variety, so that $\text{Pic}(X) \cong \mathbb{Z}^r$ and the nef cone $\text{Nef}(X)_{\mathbb{R}} \subset \text{Pic}(X)_{\mathbb{R}}$ is full-dimensional. We write $\text{Nef}(X) := \text{Nef}(X)_{\mathbb{R}} \cap \text{Pic}(X)$ for the nef monoid and $B_1, \dots, B_s$ for its unique minimal generators. For $\mathbf{v} \in \mathbb{Z}^r$, set $|\mathbf{v}| := v_1 + \dots + v_r$.

Definition 1.2. [87, Definition 6.1] *With the set-up as in the previous paragraph, if $\mathcal{F}$ is a coherent module on X , then we say that $\mathcal{F}$ is $\mathbf{d}$ -regular for $\mathbf{d} \in \mathbb{Z}^r$ if and only if for all $i > 0$:*

$$H^i(Y, \mathcal{F}(\mathbf{e})) = 0 \quad \text{for all} \quad \mathbf{e} \in L_i(\mathbf{d}) := \bigcup_{\mathbf{v} \in \mathbb{N}^s, |\mathbf{v}|=i} \left(\mathbf{d} - \sum_{j=1}^s v_j B_j + \text{Nef}(X) \right)$$

The multigraded regularity of $\mathcal{F}$ is $\text{reg}(\mathcal{F}) := \{\mathbf{d} \in \mathbb{Z}^r \mid \mathcal{F} \text{ is } \mathbf{d}\text{-regular}\}$.

One defines the multigraded regularity of a $\mathbb{Z}^r$ -graded module M over S in terms of analogous vanishing of local cohomology modules $H_B^i(M)$, where B is the irrelevant ideal of Y . The multigraded Castelnuovo–Mumford regularity is not a single number, but an infinite subset of $\mathbb{Z}^r$.

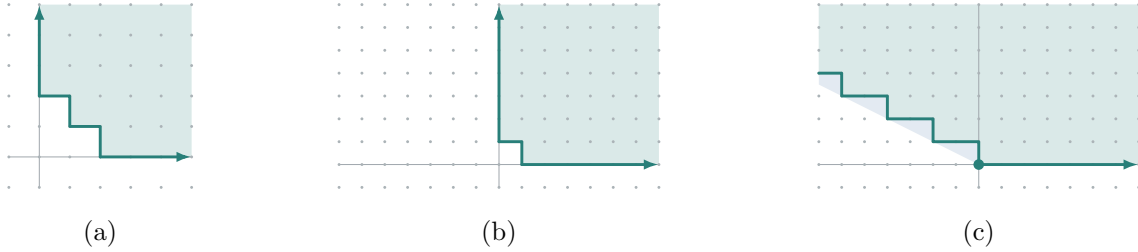


FIGURE 1. Multigraded regularity regions: (a) S/I_Z , $Z = \{([1 : 1], [1 : 4]), ([1 : 2], [1 : 5]), ([1 : 3], [1 : 6])\} \subset \mathbb{P}^1 \times \mathbb{P}^1$; (b) $\mathcal{O}_{\mathcal{H}_3}$; (c) S/I , I defining a point on $\mathcal{H}_2$.

Multigraded regularity has become a central tool in the study of uniform bounds and Hilbert schemes [88], syzygies and local cohomology [98, 18], virtual resolutions [11, 33], and asymptotic regularity [34]. Yet for more than fifteen years after its introduction, a foundational question remained open: *Is there a multigraded Eisenbud–Goto theorem?* That is, can multigraded regularity be characterized by free resolutions and truncations, as in Theorem 1.1? Direct generalizations fail even for products of projective spaces. Answering a question of MacLagan–Smith, further studied by Botbol–Chardin [18], the PI and coauthors showed that multigraded Betti numbers do not determine regularity [33, Example 5.1], ruling out an analogue of condition (2). Likewise, a $\mathbf{d}$ -regular module M need not have $M_{\geq \mathbf{d}}$ with a linear resolution, so condition (3) must also be reformulated.

A major advance came with Berkesch, Erman, and Smith’s characterization of multigraded regularity on products of projective spaces via virtual resolutions with controlled degrees [11, Theorem 2.9], a counterpart to condition (2). Building on these ideas, the PI and coauthors introduced *quasilinear resolutions* and proved a multigraded Eisenbud–Goto theorem [33]: if M is a finitely generated module over the Cox ring of a product of projective spaces with $H_B^0(M) = 0$, then M is $\mathbf{d}$ -regular if and only if $M_{\geq \mathbf{d}}$ has a quasilinear resolution generated in degree $\mathbf{d}$. This gives a precise homological characterization of multigraded regularity through resolutions of truncations.

Theorem 1.3. [33, Theorem A] *Let $\mathcal{F}$ be a coherent sheaf on $X = \mathbb{P}^{n_1} \times \dots \times \mathbb{P}^{n_r}$ and $M = \bigoplus_{\mathbf{e} \in \mathbb{Z}^r} H^0(X, \mathcal{F}(\mathbf{e}))$ the corresponding section module. For $\mathbf{d} \in \mathbb{Z}^r$ the following are equivalent:*

- (1) M is $\mathbf{d}$ -regular in the sense of [87];
- (2) M has a virtual resolution in the sense of [11]; and
- (3) $M_{\geq \mathbf{d}}$ has a quasilinear resolution in the sense of [33].

When $r = 1$, Theorem 1.3 recovers the classical Eisenbud–Goto theorem. Equally important are the methods behind the proof of Theorem 1.3: rather than working with the minimal free resolution of $M_{\geq \mathbf{d}}$ directly, we compute the graded Betti numbers of $M_{\geq \mathbf{d}}$ via a spectral sequence arising from the Fourier–Mukai transform of $\widetilde{M}$ with Beilinson’s resolution of the diagonal as kernel. This “resolution of the diagonal” approach to multigraded syzygies has become a central technique in the field [28, 26, 72, 29]. Practically, the theorem gives an effective algorithm, implemented in `VirtualResolutions` [3], for testing $\mathbf{d}$ -regularity and for producing short virtual resolutions.

1.2 Bounds & Asymptotic Behavior of Multigraded Regularity of Ideals Let X be a smooth projective toric variety with $\text{Pic}(X) \cong \mathbb{Z}^r$ and Cox ring S . We may place a partial order on $\text{Pic}(X)$ by saying that $\mathbf{d} \leq \mathbf{e}$ when $\mathbf{e} - \mathbf{d} \in \text{Nef}(X)$. Although nef twists preserve regularity, $\text{reg}(M)$ need not have a finite number of minimal elements with respect to this partial order, i.e., pictorially, the regularity region need not have finitely many corners. In [32, Example 1.1], the PI, Cranton Heller, and Sayrafi exhibit a cyclic module on a Hirzebruch surface with infinitely many minimal degrees. We show that for faithful modules such pathologies cannot occur.

Theorem 1.4. [32, Corollary 4.6] *Let M be a finitely generated graded S -module with $\text{ann}(M) = 0$ and $\widetilde{M} \neq 0$. If its homogeneous generators f_i satisfy $\mathbf{q} \leq \deg f_i$, then*

$$\text{reg}(M) \subseteq \mathbf{q} + N_X.$$

Consequently, $\text{reg}(M)$ has finitely many minimal elements.

The proof uses the fact that $\mathbf{d}$ -regularity forces $M_{\geq \mathbf{d}}$ to be generated in degree $\mathbf{d}$. A chamber and determinant argument turns a violation of the bound into a nonzero annihilator of M . This applies to every nonzero homogeneous ideal $I \subseteq S$, even though, as the aforementioned example shows, $\text{reg}(S/I)$ need not be finitely generated.

For ordinary ideal powers in the standard graded setting, Swanson established linear bounds on regularity [99], and Cutkosky–Herzog–Trung and Kodiyalam proved eventual linearity [48, 81]. Combining our bound with a resolution of the Rees algebra gives the first multigraded analog of these results, showing that regularity of powers of ideals translates at a linear rate.

Corollary 1.5. [32, Theorem 5.1] *Let $I = \langle f_1, \dots, f_s \rangle \subseteq S$ be a nonzero homogeneous ideal. There exists $\mathbf{a} \in \text{Pic}(X)$, depending only on I , such that for every integer $t > 0$ and all $\mathbf{q}_1, \mathbf{q}_2 \in \text{Pic}(X)$ satisfying $\mathbf{q}_1 \geq \deg f_i \geq \mathbf{q}_2$ for every i ,*

$$t\mathbf{q}_1 + \mathbf{a} + \text{reg}(S) \subseteq \text{reg}(I^t) \subseteq t\mathbf{q}_2 + N_X.$$

1.3 Top-Weight Cohomology of $\mathcal{A}_g$ The moduli stack of principally polarized abelian varieties of dimension g , $\mathcal{A}_g := [\mathbb{H}_g/\text{Sp}(2g, \mathbb{Z})]$, is one of the most classical objects in algebraic geometry, and yet its rational cohomology is only fully known for $g \leq 4$ [76, 70, 74, 75]. Further, until recently, beyond the stable range described by Borel [15], very little was known. In particular, Grushevsky asked whether $H^{2k+1}(\mathcal{A}_g; \mathbb{Q}) \neq 0$ for some g and k [68].

Chan, Galatius, and Payne’s computation of the top-weight cohomology of $\mathcal{M}_g$, which combines Deligne’s weight filtration with the combinatorics of tropical moduli spaces, opened a powerful new approach to the rational cohomology of moduli spaces [40] that has since become a highly active area [41, 94, 14, 37]. The PI and collaborators were the first to extend this program to $\mathcal{A}_g$. Because $\mathcal{A}_g$ lacks a modular compactification analogous to $\mathcal{M}_g \subset \overline{\mathcal{M}}_g$, this requires adapting [40] to toroidal compactifications whose boundary strata are indexed by cones in Voronoi’s reduction theory. By analyzing the combinatorics of these strata and proving an inductive relationship between the cohomology of $\mathcal{A}_g$ and that of $\mathcal{A}_{g-1}$ and $GL_g(\mathbb{Z})$, we produce a number of new cohomology classes.

Theorem 1.6. [20, Theorem A] *The rational cohomology $H^k(\mathcal{A}_g; \mathbb{Q}) \neq 0$ for:*

$$(g, k) = (5, 15), (5, 20), (6, 30), (7, 28), (7, 33), (7, 37), \text{ and } (7, 42).$$

Theorem 1.6 answers Grushevsky’s question affirmatively and, since $H^*(\mathcal{A}_g; \mathbb{Q}) \cong H^*(\mathrm{Sp}_{2g}(\mathbb{Z}); \mathbb{Q})$, produces new unstable cohomology for the symplectic group. Observations from these results inspired the subsequent construction of a Hopf algebra structure on the top-weight cohomology of $\mathcal{A}_g$ [23] and Brown’s bordification of $\mathcal{A}_g^{\mathrm{Trop}}$ [22], both of which produce further unstable cohomology of $\mathrm{GL}_n(\mathbb{Z})$. The framework of [20] is the template for the PI’s subsequent work developing tropicalization for general Shimura varieties [5], the subject of a plenary lecture at the 2025 AMS Summer Research Institute in Algebraic Geometry.

1.4 Broader Impacts from Prior NSF Support

1.4.1 Organizing. During the 2 years under consideration, I organized 5 conferences: *WAGS* (100 participants), *Gems of Combinatorics I* (40 participants), $\mathrm{Spec}(\overline{\mathbb{Q}})$ (50 participants), *BATMOBILE (2022)* (20 participants), and *Gems of Commutative Algebra II* (40 participants). In this time, I also organized special sessions at AMS Sectional Meetings. Many of these workshops, namely the *Gems of* series, focused on supporting the next generation of researchers by building research networks.

1.4.2 Mentoring. I led four reading courses with undergraduates in algebra and geometry and volunteered with the *Berkeley Math Circle* to promote mathematics to high schoolers. During Summer 2021, I led 6 undergraduates on a research project related to [20]. In Summer 2022, I advised an undergraduate student – now a math Ph.D. student with an NSF GRF – on a research project related to the work in tropical abelian varieties.

1.4.3 Virtual Mathematics. In response to the COVID-19 pandemic and the shift of many mathematical activities to virtual formats, I worked to find ways for these online activities to reach those often at the periphery. During Summer and Fall 2020, I helped with Ravi Vakil’s *Algebraic Geometry in the Time of Covid* project. This massive online open-access course in algebraic geometry brought together $\sim 2,000$ participants from around the world. In Spring 2021, I organized an 8-week virtual reading course for undergraduates in algebraic geometry and commutative algebra.

1.4.4 Community. During 2020–2023, I served on the board of *Spectra*, including as its inaugural president. I oversaw the growth and formalization of the organization, including the creation and adoption of bylaws, the creation of an invited lecture at the JMM, and a fundraising campaign that has raised over \$20,000.

2. The Geometry of Syzygies on Stacky Curves

Let X be a projective variety and L a very ample line bundle. Considering the embedding of X into projective space $X \hookrightarrow \mathbb{P}H^0(X, L) \cong \mathbb{P}^n$ defined by L , let $S_X = S/I_X$, where $S = \mathbb{C}[x_0, \dots, x_n]$ is the standard graded polynomial ring and I_X is the homogeneous defining ideal of X . The minimal graded free resolution of S_X as an S -module is an exact sequence:

$$0 \longleftarrow S_X \xleftarrow{\epsilon} F_0 \xleftarrow{d_1} \dots \xleftarrow{d_{k-1}} F_{k-1} \xleftarrow{d_k} F_k \longleftarrow \dots$$

where F_p is a graded free S -module, and the differentials satisfy some minimality conditions. Since F_p is a graded free S -module, it can be written as $\bigoplus_q S(-q)^{\beta_{p,q}(S_X)}$. The module $S(-q)$ is the ring S with a twisted grading, so that $S(-q)_d$ is equal to S_{d-q} , where S_{d-q} is the graded piece of degree $d - q$. The $\beta_{p,q}(S_X)$ are known as the *graded Betti numbers (or syzygies)* of the pair (X, L) , and we often denote them as $\beta_{p,q}(X; L)$ to indicate their dependence not just on X but on the line bundle L as well. That is to say, the syzygies depend on the extrinsic embedding of X into $\mathbb{P}^n$. An overarching theme in the study of syzygies in algebraic geometry is understanding ways that minimal graded free resolutions and graded Betti numbers also capture the intrinsic geometry of X .

Much of the work in the study of the geometry of syzygies in the last 40 years, beginning with the work of Green, has been to understand these connections. Our best understanding is for canonical

curves, i.e., when $L = K_X$, in which case Green’s conjecture [66] predicts that the Betti numbers of (X, K_X) are governed by the Clifford index of X alone:

$$\beta_{p,p+2}(X; K_X) = 0 \quad \text{if and only if} \quad p < \text{Cliff}(X). \quad (1)$$

Here $\text{Cliff}(X) := \min\{\deg(L) - 2 \dim H^0(X, L) + 2\}$, with the minimum being over all line bundles with $\deg(L) \leq g - 1$ and $\dim H^0(X, L) \geq 2$. Breakthrough work of Voisin proved this for general curves of every genus [103, 104], and in recent years new proofs have been found [4, 78]. Substantial work has gone into finding refinements and extensions of Green’s Conjecture [59, 43, 57, 58, 52, 79].

This project develops the corresponding theory for *stacky curves*, where S has a nonstandard grading and the geometry appears only after accounting for fractional divisors. Stacky curves and their section rings connect commutative algebra, geometry, and number theory [2, 36, 1, 102]. Algebraically, they arise from two-dimensional quotient singularities, and their section rings fit the Demazure–Pinkham description of normal graded rings via ample $\mathbb{Q}$ -divisors on curves [96, 51, 105]. Geometrically, Weierstrass models, hyperelliptic canonical rings, and log canonical rings all require generators of unequal degrees, placing questions about equations and syzygies in weighted projective geometry. Arithmetically, canonical rings of stacky modular and Shimura curves encode modular and automorphic forms [102], and the stack structure of modular curves drives a growing literature on rational points [13, 93, 80]. Using fractional divisors to link geometry and nonstandard gradings, the project also answers [27, Question 7.7].

2.1 Weighted Canonical Rings A stacky curve $\mathcal{X}$ is a smooth proper geometrically connected Deligne–Mumford stack of dimension one with a dense open subscheme. Concretely, we may think of $\mathcal{X}$ as a curve C together with finitely many *stacky points* carrying cyclic stabilizers [102, Lem. 5.3.10], and the coarse space map $\pi: \mathcal{X} \rightarrow C$ forgets these stabilizers. If $x_1, \dots, x_s$ are the stacky points, $e_i = \#G(x_i)$, $P_i = \pi(x_i)$, and $g = g(C)$, the *signature* of $\mathcal{X}$ is $(g; e_1, \dots, e_s)$. Following [102], the *canonical divisor* on such a stacky curve is defined to be:

$$K_{\mathcal{X}} = \pi^* K_C + \sum_i (e_i - 1) x_i = \pi^* D \quad \text{where} \quad D = K_C + \sum_i (1 - \frac{1}{e_i}) P_i \quad (2)$$

and its degree is $\deg(K_{\mathcal{X}}) = 2g - 2 + \sum_i (1 - \frac{1}{e_i})$, which may be a rational number. From this we see that the genus of $\mathcal{X}$, defined by $\deg(K_{\mathcal{X}}) = 2g_{\mathcal{X}} - 2$, is the rational number $g_{\mathcal{X}} = g + \frac{1}{2} \sum_i (1 - \frac{1}{e_i}) \geq g$. We say $\mathcal{X}$ is *hyperbolic* when $\deg(K_{\mathcal{X}}) > 0$. The *canonical ring* of $\mathcal{X}$ is

$$R(\mathcal{X}, K_{\mathcal{X}}) := \bigoplus_{d \in \mathbb{Z}} H^0(\mathcal{X}, dK_{\mathcal{X}}) \cong \bigoplus_{d \in \mathbb{Z}} H^0(C, \pi_*(dK_{\mathcal{X}})) \cong \bigoplus_{d \in \mathbb{Z}} H^0(C, \lfloor dD \rfloor)$$

where the *round down* of a $\mathbb{Q}$ -divisor $D = \sum_i a_i P_i$ on C is the honest divisor $\lfloor D \rfloor := \sum_i \lfloor a_i \rfloor P_i$, and the last isomorphism follows from [102, §5.4]. When $\mathcal{X}$ is understood, we write R for $R(\mathcal{X}, K_{\mathcal{X}})$.

This round-down is a source of an essential difference from the classical theory. Since $\lfloor K_{\mathcal{X}} \rfloor = K_C$, $R_1 = H^0(\mathcal{X}, K_{\mathcal{X}}) \cong H^0(C, K_C)$ has dimension $g(C)$ rather than $g(\mathcal{X})$. When stacky points are present, these sections vanish along the ramification divisor $K_{\mathcal{X}} - \pi^* K_C$, so they neither embed $\mathcal{X}$ nor generate R . Adding higher-degree generators yields an embedding into a weighted projective stack [19]; algebraically, R is a quotient of a polynomial ring with a nonstandard $\mathbb{Z}$ -grading.

Following Brown and Erman [27, §2], define the *minimal weighted canonical series* $\mathcal{W} = \bigoplus_{d \geq 1} \mathcal{W}_d$ by setting $\mathcal{W}_1 = R_1$ and, for $d \geq 2$, choosing $\mathcal{W}_d$ as a complement in R_d to the degree- d part of the subalgebra generated by $\mathcal{W}_1, \dots, \mathcal{W}_{d-1}$. Then $\mathcal{W}$ embeds $\mathcal{X}$ into $\mathcal{P}(\mathcal{W})$, and $R \cong S/I_{\mathcal{X}}$, where $S = \text{Sym}(\mathcal{W})$. For $g \geq 2$, Voight and Zureick-Brown’s stacky Noether–Petri theorem gives sharp degree bounds of $e := \max(1, e_1, \dots, e_s)$ for generators and $2e$ for relations, subject to exceptional cases [102, Cor. 1.4.2]. These bounds do not specify the minimal generator degrees. The first goal of this project is to close this gap by making the generation bound exact.

Let $\nu_d = \dim \mathcal{W}_d$, and record $\mathcal{W}$ by its *weight multiset*, in which each weight d appears ν_d times (we write a^m for a repeated m times). We also set $N = \sum_d \nu_d$, so that $\mathbb{P}(\mathcal{W}) = \mathbb{P}(1^{\nu_1}, 2^{\nu_2}, \dots)$ has dimension $N - 1$, together with $\sigma = \sum_d d \nu_d$ and $c = N - 2$.

Conjecture 2.1. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. The weight multiset of $\mathcal{W}$ is*

$$(1^g, (2, 3, \dots, e_1), \dots, (2, 3, \dots, e_s)),$$

with 2^{g-2} inserted if and only if C is hyperelliptic.

This conjecture implies that the weight multiset already detects whether C is hyperelliptic, and that the generation degree is exactly e . The PI hopes to prove this by producing a generator $y_{i,j} \in R_j$ of pole order exactly $j - 1$ at P_i for $2 \leq j \leq e_i$, and then showing these are minimal because $\lfloor d \frac{e-1}{e} \rfloor \leq d - 1$ forces a product of $k \geq 2$ generators of total degree j to have pole order at most $j - k$. That these elements in fact generate is the content of the following conjecture.

Filter R by the order of the poles at the stacky points by letting $F_\ell R_d := H^0(C, dK_C + \sum_i \min\{\ell, b_i(d)\} P_i)$ with $b_i(d) = \lfloor d(e_i - 1)/e_i \rfloor$. This is a multiplicative exhaustive filtration on R , and $F_0 R = R(C, K_C)$. Let $T_e = \mathbb{C}[\Gamma_e]$ for the semigroup $\Gamma_e = \{(n, \ell) : 0 \leq \ell \leq \lfloor n \frac{e-1}{e} \rfloor\}$.

Conjecture 2.2. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. The associated graded ring $\mathrm{gr}_F R$ is the fiber product of $R(C, K_C)$ with $\bigoplus_i T_{e_i}$ over morphisms given by evaluation at the P_i . In particular, there is an exact sequence*

$$0 \longrightarrow \mathrm{gr}_F R \longrightarrow R(C, K_C) \oplus \bigoplus_i T_{e_i} \longrightarrow \bigoplus_i \mathbb{C}[z_i] \longrightarrow 0, \quad (3)$$

Experiments with small curves, such as those in Figure 2, support the conjecture, but a proof remains elusive. One approach would be to show that the associated graded ring is computed by principal parts. If Conjecture 2.2 holds, the long exact sequence in Tor would supply the missing step in the proof of Conjecture 2.1. It would also relate the graded Betti numbers of $R(C, K_C)$ and $R(\mathcal{X}, K_{\mathcal{X}})$, as we explore in the following sections.

2.2 The Shape of Stacky Green's Conjecture In the classical setting, the Betti table of a canonical curve has four rows and $g - 1$ columns. Since the top and bottom rows consist of a single 1, the content of Green's conjecture lies in the two middle rows. Further, since $R(C, K_C)$ is Gorenstein, $\beta_{p,q}(C, K_C) = \beta_{g-2-p, g+1-q}(C, K_C)$. Thus, Green's conjecture (1) does two things at once: (i) determines the length of the linear strand for all canonical curves and (ii) together with the Hilbert function determines the entire Betti table of a sufficiently general curve C . Here (ii) follows from (i) since for a sufficiently general canonical curve the two rows of the Betti table of $R(C, K_C)$ do not overlap. As the following theorem shows, canonical stacky curves share some of these nice properties, being Gorenstein, but the number of rows can in general be much larger than four.

Theorem 2.3. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. Let $R = S/I$ be a minimal graded presentation of R where $S = \mathbb{K}[x_1, \dots, x_N]$ with total weight σ , and $c = N - 2$.*

- (1) *R is Gorenstein, with $\beta_{i,j}(R) = \beta_{c-i, \sigma+1-j}(R)$.*
- (2) *The weighted regularity, in the sense of [27], of R is 3.*
- (3) *Its nonzero Betti entries lie in the row range $0 \leq j - i \leq \rho - 1$, where $\rho = \sigma - N + 4$; both endpoint rows are attained. Assuming Conjecture 2.1 $g \geq 3$, we have that*

$$\rho = \begin{cases} 4 + \sum_i \binom{e_i}{2}, & C \text{ non-hyperelliptic,} \\ g + 2 + \sum_i \binom{e_i}{2}, & C \text{ hyperelliptic.} \end{cases}$$

Intervening rows are allowed to be empty.

coarse						(6; 2)							(6; 3)							
	0	1	2	3	4		0	1	2	3	4	5		0	1	2	3	4	5	6
0	1	–	–	–	–	0	1	–	–	–	–	–	0	1	–	–	–	–	–	–
1	–	6	8	3	–	1	–	6	8	3	–	–	1	–	6	8	3	–	–	–
2	–	3	8	6	–	2	–	8	24	24	8	–	2	–	8	24	24	8	–	–
3	–	–	–	–	1	3	–	–	3	8	6	–	3	–	6	24	36	24	6	–
						4	–	–	–	–	–	1	4	–	–	8	24	24	8	–
													5	–	–	–	3	8	6	–
													6	–	–	–	–	–	–	1

FIGURE 2. The Betti tables of a genus 6 trigonal canonical curve C and two stacky curves with signatures (6; 2) and (6; 3) and coarse space C .

Parts (1) and (2) follows from a direct local cohomology calculation which shows $R(\mathcal{X}, K_{\mathcal{X}})$ is a two-dimensional Cohen–Macaulay normal domain with $\mathrm{pd}_S R = c$ together with graded local duality. Alternatively, one may invokes Watanabe’s canonical module formula [105]. It is worth stressing the apartent pattern behind part (c), when C is non-hyperelliptic the coarse canonical curve gives us a Betti table that has 4 rows and $g - 1$ columns, and a stacky point of order e adds $e - 1$ columns and $\binom{e}{2}$ rows. This means that the Betti table of canonical stacky curves will have more than four rows, and the two implications of classical Green’s conjecture are best viewed as separate and new problems in the stacky world.

Research Problem 2.4. *Let $\mathcal{X}$ be a hyperbolic stacky curve with coarse space C .*

- (1) *(Weak Stacky Green’s Conjecture): Is the linear strand of $\beta(\mathcal{X}; K_{\mathcal{X}})$ determined by $\mathrm{Cliff}(C)$?*
- (2) *(Strong Stacky Green’s Conjecture): For a sufficiently general stacky curve $\mathcal{X}$, is $\beta(\mathcal{X}; K_{\mathcal{X}})$ determined by $\mathrm{Cliff}(C)$ and its Hilbert function?*

2.3 Towards a Weak Form of Stacky Green’s Conjecture To state the weak form, we need a notion of Green’s N_p -conditions adapted to non-standard gradings. Brown and Erman call $Z \subset \mathbb{P}(\mathcal{W})$ **weighted N_p** if it is normally generated and F_i is generated in degree at most w^{i+1} for all $i \leq p$, where w^j denotes the sum of the j largest weights [27, Def. 1.2]; for standard weights, this recovers Green’s classical N_p -conditions. Let $\tau_{\mathrm{BE}}(\mathcal{X})$ denote the smallest $p \geq 1$ for which Brown and Erman’s weighted N_p -condition fails. For non-stacky curves, Brown and Erman prove an analogue of Green’s N_p theorem: if a smooth curve C of genus g is embedded by a normally generated weighted linear series $\mathcal{W}$ associated to a line bundle of degree at least $2g + 1$, with $\nu_1 > g$, then C satisfies weighted N_p for all $1 \leq p \leq \dim \mathcal{W} - g - 2$ [27, Thm. 1.7]. Notice that the stacky canonical series has $\nu_1 = g$, and so one cannot hope to adapt the Brown–Erman methods directly to stacky canonical curves. Their framework is nonetheless the right one, and [27, Question 7.7] asks for precisely the canonical analogue. We predict that this threshold is determined by the length of the coarse linear strand. Set $\ell(C) := \max\{r \geq 1 : \beta_{r,r+1}(C; K_C)\}$.

Conjecture 2.5 (General Weak Stacky Green’s Conjecture). *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. Then*

$$\tau_{\mathrm{BE}}(\mathcal{X}) = (N - 2) - \ell(C).$$

Green’s conjecture predicts $\ell(C) = g - 2 - \mathrm{Cliff}(C)$, yielding $\tau_{\mathrm{BE}}(\mathcal{X}) = N - g + \mathrm{Cliff}(C)$. Thus the proposed formula in Conjecture 2.5 and Voisin’s celebrated results [103, 104] would determine the threshold for a general coarse curve of every genus $g \geq 3$, uniformly over all signatures and

choices of stacky points. The first ingredient in proving Conjecture 2.5 is a comparison of the ordinary linear strands.

Conjecture 2.6. *Let $\mathcal{X}$ be a hyperbolic stacky curve. For every $r \geq 1$, the inclusion $R(C, K_C) \hookrightarrow R$ induces an isomorphism*

$$K_{r,1}(C, K_C) \xrightarrow{\sim} K_{r,1}(\mathcal{X}, K_{\mathcal{X}}).$$

In particular, $\beta_{r,r+1}(\mathcal{X}; K_{\mathcal{X}}) = \beta_{r,r+1}(C; K_C)$.

The second ingredient to proving Conjecture 2.6 would be to locate the first failure of weighted N_p . One can hope to locate this first identifying the first failure of weighted N_p for $\text{gr}_F R$ in terms of C and the signature of $\mathcal{X}$ using the long exact sequence coming from (3). Then one may hope to transfer this failure to R via Conjecture 2.2 and upper semicontinuity.

Finally, it is worth noting that gonality and Clifford index supply no additional stacky invariant. Here Clifford index is defined using line bundles with $h^0, h^1 \geq 2$, with the usual low-genus conventions.

Proposition 2.7 (Brill–Noether rigidity). *We have $\text{Cliff}(\mathcal{X}) = \text{Cliff}(C)$ and $\text{gon}(\mathcal{X}) = \text{gon}(C)$. Every line bundle attaining either minimum is a pullback.*

Indeed, write $\mathcal{L} = \pi^* M \otimes \bigotimes_i \mathcal{T}_i^{a_i}$, where $\mathcal{T}_i = \mathcal{O}_{\mathcal{X}}(x_i)$ and $0 \leq a_i < e_i$. The round-down formula gives that $\dim H^i(\mathcal{X}, \mathcal{L}) = \dim H^i(C, M)$ and $\deg_{\mathcal{X}} \mathcal{L} = \deg M + \sum_i \frac{a_i}{e_i}$. Nonzero fractional parts increase degree without changing cohomology. This proves both equalities and forces every minimizer to have trivial stacky characters. The remaining Betti rows may require finer invariants.

2.4 Toward a Strong Form of Stacky Green’s Conjecture The strong form aims to recover the full Betti table from g , $\text{Cliff}(C)$, and the signature. The block decomposition of Conjecture 2.2 passes two checks: positive shifts for $T \neq \emptyset$ leave only $T = \emptyset$ in the linear strand, recovering Conjecture 2.6; and the maximal row is $\rho - 1$, as required by Theorem 2.3(b). The simplest stacky case admits a complete description.

Theorem 2.8 (One stacky point of order two). *Let C be non-hyperelliptic of genus $g \geq 3$, let $P \in C$, and let $\mathcal{X} = \sqrt[2]{(C, P)}$. Then for all p, q ,*

$$\beta_{p,p+q}(R) = \beta_{p,p+q}(R(C, K_C)) + \beta_{p-1,p+q-2}(R(C, K_C)) + \delta_{q,2} \binom{g-1}{p} - \varepsilon_{p,q},$$

where $\varepsilon_{p,q} = 1$ exactly for $(p, q) \in \{(1, 1), (0, 2), (g-1, 2), (g-2, 3)\}$.

Thus Problem 2.4(b) has an affirmative answer here: $\beta(R)$ is independent of P . Indeed, $\mathcal{W}$ has weights $1^g, 2$, so $N = g + 1$, $c = g - 1$, and $\rho = 5$. Conjecture 2.6 and Theorem 2.3 specify rows 0, 1, 3, 4, leaving only row 2. In each internal degree j , the Hilbert numerator determines the alternating Betti sum and hence its sole unknown entry, $\beta_{j-2,j}$; by (2), the Hilbert series depends only on the signature. For $\rho \geq 7$, at least three rows remain unknown, and these equations no longer suffice. The signature (6; 3) in Figure 2 illustrates this first unresolved range and motivates the following conjecture.

Conjecture 2.9 (Strong Stacky Green’s Conjecture). *Let C be a smooth non-hyperelliptic curve of genus $g \geq 3$. The Betti numbers $\beta_{p,q}(\mathcal{X}; K_{\mathcal{X}})$ depends only on C and the signature of $\mathcal{X}$, and not on the position of the stacky points.*

Voight and Zureick-Brown show that the Hilbert series and the generic initial ideal of $I_{\mathcal{X}}$ depend only on the signature [102]. Any dependence on position is therefore invisible to both and, assuming Conjecture 2.6 and Theorem 2.3, is confined to the middle rows. The decisive test is a stacky point of order $e \geq 3$ at a general point of C , where the evaluation maps in (3) detect canonical jets. Proposition 2.7 forces any invariant governing the middle rows to be local at the marked points rather than global on line bundles. This points toward pointed Brill–Noether theory with

prescribed ramification, and toward limit linear series and tropical methods for imposing ramification conditions [95, 85, 77, 45, 84]. Resolving the case of a single point of order three is the central goal of this part of the proposal. If true Conjecture 2.9 naturally leads to the following goal.

Research Problem 2.10. *Derive a closed formula for the Betti numbers $\beta_{p,q}(\mathcal{X}; K_{\mathcal{X}})$, in terms of the Betti numbers of (C, K_C) and the signature of $\mathcal{X}$.*

2.5 Beyond Stacky Canonical Curves The desired proof of Conjecture 2.6 seems to never truly use that the line bundle of interest is $K_{\mathcal{X}}$, and so we hope it will generalize to any divisor on a stacky curve of the form $\mathcal{D} = \pi^*L \otimes \bigotimes_i \mathcal{T}_i^{a_i}$, where L is a very ample line bundle on C and $\mathcal{T}_i := \mathcal{O}_{\mathcal{X}}(x_i)$ is the line bundle of the reduced stacky point, so that $\deg \mathcal{T}_i = 1/e_i$ and $\mathcal{T}_i^{e_i} \cong \pi^*\mathcal{O}_C(P_i)$. Here $\ell(C)$ should be replaced by the Koszul cohomology invariant of $[\mathcal{D}]$, and this would answer [27, Question 7.6]. With M. Banks, C. Manon, and P. Wei, the PI is pursuing the complementary high-degree regime: we have built the multigraded kernel bundles whose cohomology computes the weighted Betti numbers of $R(\mathcal{X}, \mathcal{L})$, and are proving the vanishing theorems that would give weighted N_p with p linear in $\deg \mathcal{L}$ for $g \leq 1$. Concretely, we expect the following.

Conjecture 2.11. *Let $\mathcal{X}$ be a hyperbolic stacky curve of signature $(g; e_1, \dots, e_s)$ and let $\mathcal{D}$ be as above with $\deg[\mathcal{D}] \geq 2g + 1$. Then*

$$\tau_{\text{BE}}(\mathcal{X}, \mathcal{D}) = \deg[\mathcal{D}] - 2g - 1 + \sum_i (e_i - 1),$$

so that $R(\mathcal{X}, \mathcal{D})$ satisfies weighted N_p for all p below this bound.

This is the exact analogue of the classical statement for line bundles of degree at least $2g + 1$, shifted by the $\sum_i (e_i - 1)$ extra generators, and it agrees with Conjecture 2.5 in the canonical limit. A further target is the *log* canonical ring $R(\mathcal{X}, K_{\mathcal{X}} + \Delta)$ with $\deg \Delta > 0$, the ring of modular forms for $\Gamma_0(N)$ and the application motivating [102]. Here $K_{\mathcal{X}} + \Delta$ is not dualizing and Theorem 2.3 fails, but Watanabe’s formula identifies ω_R with the section module of the residual divisor, so duality should become a pairing between $R(\mathcal{X}, K_{\mathcal{X}} \pm \Delta)$; we expect the weak form to survive verbatim and Conjecture 2.5 not to. All of this sits in a fast-moving neighborhood: beyond [27, 24, 7], Brown and Erman have studied positivity and nonstandard graded Betti numbers [25], Martinova has begun the asymptotic theory of syzygies of weighted projective spaces [89], and Koszulness for truncations of nonstandard graded rings is now understood [50].

3. Multigraded Castelnuovo–Mumford Regularity: Bounds and Behavior

As discussed in Section 1.1, Castelnuovo–Mumford regularity links cohomological vanishing to the degrees of equations and syzygies. The PI’s work extends this link to products of projective spaces and proves finite-generation and asymptotic results for regularity on toric varieties [32, 33]. Basic questions remain open: regularity can have infinitely many corners, it is not determined by the multigraded Betti numbers, and few geometric bounds are known. This project aims to explain the shape of the regularity region through the geometry of its support, bound its corners using syzygies, and bound the regularity of curves using intersection theory.

Throughout this section, X is a smooth projective toric variety of dimension n over an algebraically closed field $\mathbb{K}$ of characteristic zero, with Cox ring S , irrelevant ideal B , and $\text{Pic}(X) \cong \mathbb{Z}^r$. We fix the minimal generators $B_1, \dots, B_s$ of the nef monoid $\text{Nef}(X)$ and use these to define regularity. A *corner* is a minimal element with respect to the partial order induced by $\text{Nef}(X)$.

3.1 The Shape and Minimal Degrees of Multigraded Regularity Over a standard graded polynomial ring, a nonzero finitely generated graded module M has regularity region $m + \mathbb{N}$, where $m = \max\{q - p \mid \beta_{p,q}(M) \neq 0\}$. Thus its Betti table determines a single corner. In several gradings, neither conclusion survives [33, 32]. This leads to the following questions.

Research Problem 3.1. (1) *How does the geometry of the support govern the infinite directions of regularity, and when can the region be described by finite data?*
(2) *How do syzygies constrain the location and number of corners of multigraded regularity?*

The diagonal $\Delta \subset \mathbb{P}^1 \times \mathbb{P}^1$ illustrates the first question. Since $\mathcal{O}_\Delta(a, b) \cong \mathcal{O}_{\mathbb{P}^1}(a + b)$, one readily checks that $\text{reg}(\mathcal{O}_\Delta) = \{(a, b) \in \mathbb{Z}^2 \mid a + b \geq 0\}$ and hence has a corner at $(a, -a)$ for all $a \in \mathbb{Z}$. The feature that causes this is that a divisor on $\mathbb{P}^1 \times \mathbb{P}^1$ that is not nef can become nef when restricted to Δ . Here the restriction of $\mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^1}(a, b)$ has degree $a + b$, so the relevant nef region is a half-plane rather than a quadrant. More generally, Fujita vanishing on the support suggests that these additional nef directions should govern the large-scale shape of regularity.

To describe such behavior, set $P_{\mathcal{F}} := \overline{\text{conv}(\text{reg}(\mathcal{F}))}$ for a nonzero coherent sheaf $\mathcal{F}$. Its *recession cone* consists of vectors $\mathbf{v}$ such that $P_{\mathcal{F}} + t\mathbf{v} \subseteq P_{\mathcal{F}}$ for every $t \geq 0$. This records the directions in which one can move indefinitely from every point of the closed convex region. Nef persistence gives $\text{Nef}(X) \subset \text{rec}(P_{\mathcal{F}})$, but the diagonal shows that the inclusion can be strict. For a sheaf with support Z , this suggests $C_Z = \{\mathbf{d} \in \text{Pic}(X)_{\mathbb{R}} \mid \mathbf{d}|_Z \text{ is nef}\}$.

Conjecture 3.2. *If $\mathcal{F}$ is a nonzero coherent sheaf on X and $Z = \text{supp}(\mathcal{F})$, then the unbounded directions of the regularity of $\mathcal{F}$ are dual to the curve classes carried by the support:*

$$\text{rec}(P_{\mathcal{F}}) = \text{cone}(C_Z).$$

I hope to prove Conjecture 3.2 by proving the stronger claim: there exist $\mathbf{a}, \mathbf{b} \in \text{Pic}(X)$ with

$$\mathbf{a} + (C_Z \cap \text{Pic}(X)) \subset \text{reg}(\mathcal{F}) \subset \mathbf{b} + C_Z. \quad (4)$$

Fujita vanishing gives the first inclusion: after one sufficiently positive ample twist, the finitely many sheaves appearing in the regularity tests have no higher cohomology after any support-nef twist. For the reverse inclusion, I will use the fact that regularity implies global generation [87]. A fixed presentation of $\mathcal{F}$ restricts to each normalized support curve and bounds the degree of its torsion-free quotient in terms of an ample class. Global generation after a regular twist should then give a uniform lower bound on intersection with every such curve. The objective is to make this comparison effective, so that the support determines both the asymptotic cone and a bounded range in which its boundary can vary.

Conjecture 3.2 also provides a multigraded analog of the fact that a nonzero coherent sheaf on $\mathbb{P}^n$ has regularity $-\infty$ precisely when it is supported on finitely many points. However, as is often the case, the answer can take different forms.

Corollary 3.3 (Regularity in every degree). *Assume Conjecture 3.2. If $\mathcal{F}$ is a nonzero coherent sheaf on X and $Z = \text{supp}(\mathcal{F})$, then:*

- (1) $\text{reg}(\mathcal{F}) = \text{Pic}(X)$ if and only if $\dim Z = 0$.
- (2) $\text{reg}(\mathcal{F}) \not\subset \mathbf{d} + \text{Nef}_{\mathbb{R}}(X)$ for all $\mathbf{d} \in \text{Pic}(X)$ if and only if $C_Z \not\supseteq \text{Nef}(X)$.

The same bounds suggest when infinitely many ambient corners can be replaced by finitely many degrees on the support. Let $\rho : \text{Pic}(X) \rightarrow \text{Pic}(Z)$ be the restriction map $L \mapsto L|_Z$. Twists differing by $\ker(\rho)$ give identical cohomology tests, so regularity is invariant under $\ker(\rho)$.

The uniform curve bound also gives a finite reduction. Write $\mathcal{F} = i_*\mathcal{G}$, let $\rho : \text{Pic}(X) \rightarrow \Lambda_Z = \text{im}(\text{Pic}(X) \rightarrow \text{Pic}(Z))$, and put $P_Z = \rho(N_X)$. Define $R_Z(\mathcal{G})$ using the unchanged restricted list i^*B_j : membership of A requires $H^q(Z, \mathcal{G} \otimes A \otimes \bigotimes_j (i^*B_j)^{-v_j}) = 0$ for $1 \leq q \leq \dim Z$, $\mathbf{v} \in \mathbb{N}^s$, and $|\mathbf{v}| = q$. The projection formula gives $\text{reg}(\mathcal{F}) = \rho^{-1}(R_Z(\mathcal{G}))$.

Corollary 3.4 (Finite reduction to the support). *Assume Conjecture 3.2. There are finitely many degrees $\mathbf{d}_1, \dots, \mathbf{d}_t$ with*

$$\text{reg}(\mathcal{F}) = \bigcup_{j=1}^t (\mathbf{d}_j + \text{Nef}(X) + \ker(\rho)) \quad \text{if and only if} \quad C_Z = \text{Nef}(X) + \ker(\rho)_{\mathbb{R}}.$$

The condition says that every support-nef direction comes from an ambient nef direction after allowing twists trivial on the support. The outer bound in (4) and finite generation of lattice points in rational polyhedral cones then reduce regularity to finitely many degrees modulo $\ker(\rho)$. For the diagonal, $\ker(\rho) = \mathbb{Z}(1, -1)$ and the single degree $(0, 0)$ suffices. Finding these degrees effectively is the next step.

Turning to the second question, my work with Heller and Sayrafi proves finite generation for faithful modules [32]. I conjecture a bound uniform over the minimal regularity elements for a fixed Betti table.

Conjecture 3.5. *For fixed X , there is an explicitly computable finite lattice box $Q_X(\beta)$ containing every corner of $\text{reg}(M)$ for all finitely generated faithful graded modules M with $\beta(M) = \beta$ and $\widetilde{M} \neq 0$. On $X = \prod_{j=1}^r \mathbb{P}^{n_j}$, $n_j \geq 1$, faithfulness can be replaced by $H_B^0(M) = 0$, and one can take*

$$Q_X(\beta) = \prod_{j=1}^r [\ell_j(\beta), U_j(\beta)], \quad \ell_j(\beta) = \min\{a_j \mid \beta_{0,\mathbf{a}} \neq 0\}.$$

In joint work with Lauren Cranton Heller, we have proved this conjecture for $\mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$. The key is to bound where the regularity region stabilizes in each coordinate. Above a threshold T_y determined by the Betti table, membership of (a, b) depends only on whether $a \geq r_x(M)$; there is a symmetric statement beyond a threshold T_x . Thus every corner lies in a finite rectangle. We obtain these thresholds by relating projection cohomology to the regularity of graded slices and bounding the relevant partial Koszul homology using the shifts and ranks of the free resolution. We are extending the argument to arbitrary products by induction along projections. For general toric varieties, where the nef cone need not be a coordinate orthant, I will investigate how resolutions of the diagonal encode the corresponding stabilization [72, 29].

Corollary 3.6. *Assume Conjecture 3.5 for $X = \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_r}$. Among finitely generated graded modules with fixed multigraded Betti table β :*

- (1) *Only finitely many regularity regions occur.*
- (2) *Each region can be computed by testing the finitely many lattice degrees in $Q_X(\beta)$ using the quasilinear-truncation criterion of [33].*
- (3) *If every side of $Q_X(\beta)$ has length at most w , each region has at most $(w + 1)^{r-1}$ corners.*

Indeed, the regular degrees in the box generate the whole region under addition by $\mathbb{N}^r$. Its corners form an antichain, and deleting any one coordinate gives an injective projection of that antichain. Thus Picard rank controls the exponent in the corner bound. For powers I^p , boxes of width $O(p)$ would give at most $O(p^{r-1})$ corners, a quantitative extension of the asymptotic bounds in Section 1.

3.2 A Gruson, Lazarsfeld, and Peskine Style Bound for Multigraded Regularity A foundational result of Gruson, Lazarsfeld, and Peskine [69] states that if $C \subset \mathbb{P}_{\mathbb{K}}^r$ is an integral nondegenerate curve defined over an algebraically closed field $\mathbb{K}$, then

$$\text{reg}(I_C) \leq \deg(C) - r + 2$$

where $I_C \subset \mathbb{K}[x_0, \dots, x_r]$ is the (saturated) homogeneous defining ideal of C . Since $\deg(C) = C \cdot H$ for the hyperplane class H , this theorem provides a remarkably strong bridge between the elementary intersection theory of C and the degrees of its defining equations and syzygies. It sharpens and completes classical work of Castelnuovo [38] on space curves, and it settles the curve case of the bound $\text{reg}(I_C) \leq \deg(C) - \text{codim}(C) + 1$ conjectured by Eisenbud and Goto [55]. Beyond its original statement, [69] has inspired decades of work on geometric regularity bounds: finding optimal bounds [97, 86, 82, 83], and bounding regularity by the degrees of defining equations [12, 53, 49]. Recent counterexamples to the Eisenbud–Goto conjecture in general [90, 39, 91] make such genuinely

geometric bounds all the more valuable. I propose to establish a multigraded analogue of the Gruson–Lazarsfeld–Peskin theorem for curves in smooth projective toric varieties.

Research Problem 3.7. *Prove a multigraded analogue of Gruson, Lazarsfeld, and Peskin’s theorem for curves in smooth projective toric varieties.*

The objective is an explicit subset of $\text{reg}(\mathcal{I}_C)$ whose shape reflects the divisor relations of the ambient variety [87]. Cobb has made progress toward Problem 3.7 by adapting the kernel bundle techniques of Gruson, Lazarsfeld, and Peskin to products of projective spaces [44]. The shape of my proposed answer comes from how the classical bound is proved. By definition, $\mathcal{I}_C$ is m -regular if $H^q(\mathcal{I}_C(m - q)) = 0$ for $q \geq 1$. The exact sequence

$$0 \longrightarrow \mathcal{I}_C \longrightarrow \mathcal{O}_{\mathbb{P}^r} \longrightarrow \mathcal{O}_C \longrightarrow 0$$

splits the needed vanishing into three separate questions: (i) Ambient ($q \geq 3$): Since C is a curve, $H^q(C, \mathcal{O}_C(m - q)) = 0$ for $q \geq 2$, and the exact sequence gives $H^q(\mathbb{P}^r, \mathcal{I}_C(m - q)) \cong H^q(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(m - q))$ for $m \geq 3$. Projective space has no intermediate cohomology, and its top cohomology vanishes at these twists. Thus only the $q = 1, 2$ cases remain. (ii) Speciality ($q = 2$): When $q = 2$, we have that $H^2(\mathcal{I}_C(m - 2)) \cong H^1(\mathcal{O}_C(m - 2))$, and this reduces to whether or not $\mathcal{O}_C(m - 2)$ is non-special. The naive Riemann–Roch computation shows this is true if $(m - 2) \deg(C) > 2p_a(C) - 2$. When $m = \deg(C) - r + 2$, this follows from Castelnuovo’s genus bound. (iii) Interpolation ($q = 1$): The group $H^1(\mathbb{P}^r, \mathcal{I}_C(m - 1))$ is the cokernel of the restriction map $H^0(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(m - 1)) \rightarrow H^0(C, \mathcal{O}_C(m - 1))$. Its vanishing means that every section on C extends to an ambient polynomial. Establishing this surjectivity is the heart of the theorem and is addressed through kernel bundle arguments.

The constant $\deg(C) - r + 2$ also has a geometric explanation. If ℓ is a secant line of C , the nondegeneracy of C forces the length of $C \cap \ell$ to be at most $\deg(C) - r + 2$. Using a speciality and interpolation argument involving ℓ , we see that an s -secant line forces $\text{reg}(\mathcal{I}_C) \geq s$. Hence the Gruson, Lazarsfeld, and Peskin bound arises from a maximal secant line to C .

Let X be a smooth projective toric variety of dimension $n \geq 2$ over an algebraically closed field of characteristic zero, and $B_1, \dots, B_s$ the minimal generators of $\text{Nef}(X)$. Say that an integral curve $C \subset X$ is *nondegenerate* if each restriction $H^0(X, B_j) \rightarrow H^0(C, B_j|_C)$ is injective. Set $r_j := \dim_{\mathbb{K}} H^0(X, B_j) - 1$, $d_j := \langle B_j \cdot C \rangle$, and the *defect* $\delta_j := d_j - r_j$. MacLagan–Smith regularity of $\mathcal{I}_C$ at $\mathbf{d}$ tests the twists $\mathbf{d} - B_i$, then $\mathbf{d} - B_i - B_j$, and so on, so the same three questions arise. I propose answering each by a region depending only on the numerical class C and on X :

- (1) *Ambient vanishing.* Membership in $\text{reg}(\mathcal{O}_X)$ gives $H^q(X, \mathcal{O}_X(\mathbf{d} - \sum_j v_j B_j)) = 0$ for $q > 0$ and $|\mathbf{v}| = q$. Since C is a curve, for $q \geq 3$ these are exactly the ideal-sheaf tests.
- (2) *Nonspeciality.* Choose an explicit numerical bound $\widehat{G}_X([C]) \geq p_a(C)$. For example, $A = \sum_j B_j$ is very ample, and multiplication of the restricted section spaces shows that C spans at least a $\mathbb{P}^R$, where $R = \sum_j r_j$. Thus one may take Castelnuovo’s bound $\pi(D, R)$, where $D = \sum_j d_j$: writing $D - 1 = m(R - 1) + \epsilon$ with $0 \leq \epsilon < R - 1$, this is $\binom{m}{2}(R - 1) + m\epsilon$. On a surface, adjunction gives the exact genus and can be used instead. Every second twist is nonspecial on

$$\mathcal{H}_X([C]) := \left\{ \mathbf{d} \in \text{Pic}(X) \mid \mathbf{d} \cdot C \geq 2\widehat{G}_X([C]) - 1 + 2 \max_j d_j \right\},$$

because $\deg \mathcal{O}_C(\mathbf{d} - B_i - B_j) > 2p_a(C) - 2$.

- (3) *Interpolation.* A line meeting a projective curve properly in a scheme of length k forces its ideal to have regularity at least k . Nondegeneracy bounds this length by $\deg(C) - N + 2$, suggesting the classical threshold. To formulate a toric analogue, set $M_X := \overline{\text{NE}}(X) \cap N_1(X)_{\mathbb{Z}}$ and $e_j(\gamma) := B_j \cdot \gamma$.

Call $0 \neq \gamma \in M_X$ *eligible* if $e_j(\gamma) < r_j$ for some j , and put

$$q_C(\gamma) := \min_{j: e_j(\gamma) < r_j} (\delta_j + e_j(\gamma)), \quad w(\gamma) := \max_j e_j(\gamma).$$

If an integral rational curve $R \neq C$ has eligible class γ , its image under B_j spans at most $\mathbb{P}^{e_j(\gamma)}$. The resulting sections vanishing on R , together with nondegeneracy of C , give $\text{length}(C \cap R) \leq q_C(\gamma) + 1$. When R is smooth, arbitrary values on a subscheme of length $q_C(\gamma) + 1$ can be interpolated by a line bundle of degree at least $q_C(\gamma)$. Requiring this degree at every first twist motivates the candidate below.

$$\mathcal{S}_X([C]) := \{\mathbf{d} \in \text{Pic}(X) \mid \mathbf{d} \cdot \gamma \geq q_C(\gamma) + w(\gamma) \text{ for every eligible } \gamma \in M_X\}.$$

The definition tests all eligible lattice classes, including nonprimitive ones; the claim that these numerical inequalities suffice for interpolation is conjectural.

Conjecture 3.8. *If $C \subset X$ is an integral nondegenerate curve, then*

$$\mathcal{R}_X([C]) := \mathcal{S}_X([C]) \cap \mathcal{H}_X([C]) \cap \text{reg}(\mathcal{O}_X) \subseteq \text{reg}(\mathcal{I}_C).$$

On $\mathbb{P}^r$, the line class gives $m \geq \deg(C) - r + 2$; the higher eligible multiples give weaker inequalities, and Castelnuovo's genus bound supplies nonspeciality at the same threshold. Thus the conjecture recovers the classical theorem. The proposed bound can improve on Cobb's bound even for a small, geometrically familiar curve. Consider

$$C = \{([u^3 : u^2v : v^3], [u^2 : uv : v^2]) \mid [u : v] \in \mathbb{P}^1\} \subset X = \mathbb{P}^2 \times \mathbb{P}^2.$$

This pairs the normalization of a cuspidal plane cubic with the conic embedding; in particular, C is smooth. Its multidegree is $(3, 2)$ and its defects are $(1, 0)$. The inequalities defining $\mathcal{S}_X([C])$ reduce to $a \geq 1$ and $b \geq 2$, while the genus estimate $\widehat{G}_X([C]) = \pi(5, 4) = 1$ gives the redundant condition $3a + 2b \geq 7$. In fact,

$$\mathcal{R}_X([C]) = \text{reg}(\mathcal{I}_C) = (1, 2) + \mathbb{N}^2,$$

whereas Cobb's bound gives $(3, 3) + \mathbb{N}^2$ [44, Proposition B]. Thus the proposed bound is sharp in this example and improves both coordinates of Cobb's corner. I have proved the conjecture for curves in $\mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$ with a minimal-degree projection, and computations for curves in $\text{Bl}_p \mathbb{P}^3$ have found no counterexamples. The remaining condition is precise: for every $\mathbf{d} \in \mathcal{R}_X([C])$, prove that

$$H^0(X, \mathcal{O}_X(\mathbf{d} - B_i)) \longrightarrow H^0(C, \mathcal{O}_C(\mathbf{d} - B_i))$$

is surjective for each i . The ambient and nonspeciality conditions already give the other regularity tests. For general X , I will combine the classical use of an auxiliary line bundle on the normalization with the resolution of the diagonal of Hanlon, Hicks, and Lazarev [69, 44, 72]. Write its terms as $\mathcal{D}_p = \bigoplus_\lambda \mathcal{O}_X(-A_{p\lambda}) \boxtimes \mathcal{O}_X(-E_{p\lambda})$. For the normalization map $f: \widetilde{C} \rightarrow X$ and a line bundle L on $\widetilde{C}$, pullback, tensor product, and pushforward give a spectral sequence with terms

$$E_1^{-p,q} = \bigoplus_\lambda H^q(\widetilde{C}, L \otimes f^* \mathcal{O}_X(-A_{p\lambda})) \otimes \mathcal{O}_X(-E_{p\lambda}), \quad q = 0, 1,$$

converging to $f_* L$ in total degree zero. The task is to bound the choice of L that kills the relevant H^1 groups, then control the resulting presentation and its determinantal equations using the inequalities defining $\mathcal{R}_X([C])$. Singular curves also require control of the finite-length discrepancy introduced by normalization. I will begin with Brown and Sayrafi's explicit resolution in Picard rank two [29], where $\text{Bl}_p \mathbb{P}^3$ provides a concrete test; curves in $\mathbb{P}^1 \times \mathbb{P}^2$ provide a first project for a graduate student. A proof would give intersection-theoretic bounds for equations in Cox rings [87] and, on products of projective spaces, for the quasilinear truncations in Theorem 1.3.

3.3 Multigraded Regularity of Torus-Invariant Subvarieties On a smooth projective toric variety $X = X_\Sigma$, the torus-invariant irreducible subvarieties are in dimension-reversing bijection with the cones of Σ . If $S = \mathbb{K}[x_\rho \mid \rho \in \Sigma(1)]$ is the Cox ring, the subvariety Z_σ corresponding to $\sigma \in \Sigma$ has Cox ideal $I_{Z_\sigma} = \langle x_\rho \mid \rho \in \sigma(1) \rangle$, which is resolved by a multigraded Koszul complex. Even so, the multigraded regularity of these subvarieties can be subtle: the four torus-fixed points of the Hirzebruch surface $\mathcal{H}_2$ exhibit two distinct regularities, one with infinitely many minimal corners [32]. Thus, even these highly structured subvarieties provide an interesting testing ground for understanding how multigraded regularity reflects positivity beyond the information provided by the syzygies. As of yet, there are now concrete descriptions of the multigraded regularity of torus-invariant points, which is a question I seek to address in this project.

Research Problem 3.9. *Let $Z_\sigma \subset X$ be an irreducible torus-invariant subvariety corresponding to the cone σ . Give an explicit combinatorial characterization of $\text{reg}(I_{Z_\sigma})$ and $\text{reg}(S/I_{Z_\sigma})$.*

Note that this question asks for the regularity of both I_{Z_σ} and S/I_{Z_σ} because in the multigraded setting, unlike in the standard graded setting, the fact that $\text{reg}(S)$ need not have a single corner means they can behave quite differently. Again, see [32].

Our approach to Problem 3.9 combines the combinatorics of Hilbert functions with an explicit description of the higher cohomology of line bundles restricted to torus-invariant subvarieties. With my graduate student Jacob Lehman Duke, we have settled the case of torus-fixed points. For a maximal cone σ , the variables surviving in S/I_{Z_σ} are those indexed by rays outside σ , so the degrees of monomials not vanishing at Z_σ form the monoid

$$Q_\sigma = \left\{ \sum_{\rho \notin \sigma(1)} a_\rho [D_\rho] \mid a_\rho \in \mathbb{N} \right\} \subset \text{Pic}(X).$$

By smoothness, the classes $[D_\rho]$ for $\rho \notin \sigma(1)$ form a lattice basis of $\text{Pic}(X)$, so the Hilbert function of S/I_{Z_σ} is 1 on Q_σ and 0 elsewhere.

Proposition 3.10. *If $Z_\sigma \in X$ is a torus-fixed point corresponding to a maximal cone $\sigma \in \Sigma$, then*

$$\text{reg}(S/I_{Z_\sigma}) = Q_\sigma \quad \text{and} \quad \text{reg}(I_{Z_\sigma}) = \text{reg}(S) \cap \left(\bigcap_{i=1}^t (C_i + Q_\sigma) \right).$$

Since S/I_{Z_σ} has no higher sheaf cohomology, its regularity is the set of degrees where its Hilbert function equals its Hilbert polynomial [87, Proposition 6.4], namely Q_σ . A cohomology argument then reduces $\text{reg}(I_{Z_\sigma})$ to surjectivity of evaluation at each first twist, i.e. $\mathbf{d} - B_j \in Q_\sigma$ for all j , giving the intersection. Notably, $\text{reg}(S/I_{Z_\sigma})$ depends only on the rays of Σ and on σ , so it is unchanged across toric models sharing these data. For positive-dimensional torus-invariant subvarieties, a similar monomial count gives the Hilbert function of S/I_{Z_σ} , leaving the higher cohomology of line bundles restricted to Z_σ . Since Z_σ is toric, with fan given by the images of the cones containing σ modulo σ , we plan to compute these vanishing regions inductively via toric Čech complexes.

Such descriptions would also make existing degeneration bounds explicit. Maclagan and Smith bound the regularity of a monomial quotient by intersecting shifted regularity regions of the factors in a Stanley filtration, which for saturated monomial ideals are coordinate rings of orbit closures [88, Theorem 4.1]. A combinatorial answer to Problem 3.9 would compute these ingredients directly from the fan, and the resulting bounds pass from initial ideals to the original quotient by semicontinuity [87, Proposition 6.13].

4. Uniform Homological Algebra: Regularity on the Cox Category

This part of the proposal is deliberately speculative: *multigraded regularity may be asking the right questions, but in the wrong category.* Work of the PI and others, exposes the fragility of the classical link between cohomological vanishing and syzygies [87, 33, 32]. Even the Cox ring of the

Hirzebruch surface $\mathcal{H}_2$, as a free module over itself, is not $\mathbf{0}$ -regular [87, Example 4.3]. I propose to test whether treating birational models together provides a more natural setting.

Uniformity across toric birational models is a growing theme in multigraded algebra, beginning with Hanlon–Hicks–Lazarev’s diagonal, which depends only on the fan rays [72]. Work on short resolutions, cellular constructions, and derived categories has developed this theme [26, 60, 61, 62, 47, 16, 10, 71]. Ballard et al. give a striking reformulation of King’s conjecture: a full strong exceptional collection of line bundles can fail to exist on an individual smooth projective toric variety, yet exists in the *Cox category* incorporating its birational models [6]. I propose to use its uniform diagonal to “repair” classical regularity and make the structure sheaf $\mathbf{0}$ -regular on every smooth projective model.

4.1 The Cox Category and the Diagonal Let X be a projective normal toric variety over an algebraically closed field $\mathbb{K}$ of characteristic zero, with Cox ring S . Varying the irrelevant ideal gives models X_i indexed by maximal GKZ chambers and associated smooth toric stacks $\mathcal{X}_i$, with $\mathcal{X}_i = X_i$ when X_i is smooth. Let $D^b(-) := D^b(\text{Coh}(-))$ and all functor that follow be derived. Following Ballard et al. and choose a common smooth toric stack with proper birational maps $\pi_i: \tilde{\mathcal{X}} \rightarrow \mathcal{X}_i$ and set $D_{\text{Cox}}(X) := \langle \pi_i^* D^b(\mathcal{X}_i) \rangle \subset D^b(\tilde{\mathcal{X}})$. This category is independent of the choice and contains each $D^b(\mathcal{X}_i)$ as a full subcategory. The Cox grading determines a half-open zonotope:

$$Q: \mathbb{Z}^{\Sigma(1)} \rightarrow \text{Cl}(X), \quad Z_X := Q_{\mathbb{R}} \left((-1, 0]^{\Sigma(1)} \right).$$

The classes mapping into Z_X form the finite *Bondal–Thomsen collection* $\Theta \subset \text{Cl}(X)$. For $\alpha \in \Theta$, pull back its line bundle from a chamber containing $-\alpha$. The resulting objects $\mathcal{O}_{\text{Cox}}(\alpha)$ are independent of this choice and form a full strong exceptional collection [6].

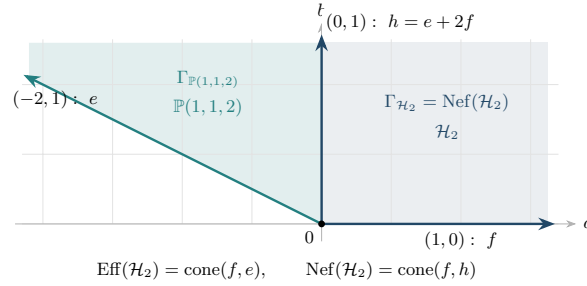


FIGURE 3. GKZ fan of $\mathcal{H}_2$. The second chamber gives $\mathbb{P}(1, 1, 2)$, whose coarse space is obtained by contracting the (-2) -section e .

The uniform Hanlon–Hicks–Lazarev (HHL) complex $H_{S,S}$ lifts to a diagonal kernel inducing the identity on $D_{\text{Cox}}(X)$ [6], converting cohomology into terms of a complex as in the PI’s work on quasilinear truncations. With the first factor as input, minimize $H_{S,S}$, and write

$$(H_{S,S}^{\min})_p = \bigoplus_{\alpha, \beta \in \Theta} (S(\alpha) \boxtimes S(\beta))^{m_{p, \alpha, \beta}}, \quad p \geq 0,$$

with $p = 0$ adjacent to the augmentation. For the occurring input labels $\alpha \in \Theta_H$, define

$$h(\alpha) := \min\{p : m_{p, \alpha, \beta} \neq 0 \text{ for some } \beta\}, \quad \nu(\alpha) := \max\{1, h(\alpha)\}.$$

The *height* is a label’s first homological degree in the minimal HHL complex, not necessarily its face codimension in the zonotope. The HHL periodic hyperplane arrangement supplies the combinatorics.

Research Problem 4.1. Describe $h(\alpha)$ for every $\alpha \in \Theta_H$ combinatorially from the divisor-class map, zonotope, and GKZ fan, without constructing and minimizing the entire diagonal complex.

I will use Favero–Sapronov’s stratified-torus methods and Han’s minimal-resolution constructions [62, 71] to relate heights to Cox degrees and GKZ walls. A first test is Figure 4: two labels of different heights become the same line bundle on the weighted projective model.

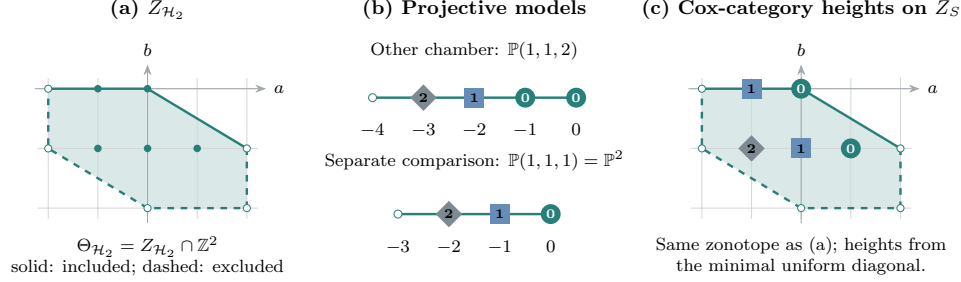


FIGURE 4. Zonotopes and heights [47, §4], with $Z_{H_2} = Z_S$ in panels (a),(c). Filled markers are included; open circles mark excluded vertices.

4.2 A Candidate Regularity Theory

Definition 4.2 (Proposed Cox regularity). *An object $\mathcal{E} \in D_{\text{Cox}}(X)$ is $\mathbf{0}$ -regular if*

$$\text{Ext}_{D_{\text{Cox}}(X)}^q(\mathcal{O}_{\text{Cox}}(\alpha)^\vee, \mathcal{E}) = 0 \quad \text{for all } \alpha \in \Theta_H \text{ and } q \geq \nu(\alpha).$$

For a coherent sheaf $\mathcal{F}$ on $\mathcal{X}_i$, the Θ -Transform Lemma and projection formula give

$$\text{Ext}^q(\mathcal{O}_{\text{Cox}}(\alpha)^\vee, \pi_i^*(\mathcal{F}(\mathbf{d}))) \cong H^q(\mathcal{X}_i, \mathcal{F}(\mathbf{d} + \alpha)).$$

Let $\mathcal{R}_i(\mathcal{F})$ be the classes $\mathbf{d} \in \text{Pic}(\mathcal{X}_i)$ satisfying these vanishings. Bondal–Thomsen vanishing makes $\mathcal{O}_{\text{Cox}}$ $\mathbf{0}$ -regular and gives $\text{Nef}(\mathcal{X}_i) \cap \text{Pic}(\mathcal{X}_i) \subseteq \mathcal{R}_i(\mathcal{O}_{\mathcal{X}_i})$.

The cutoff reflects diagonal reconstruction: degree- q cohomology in diagonal degree p contributes in position $p - q$. Requiring $q < h(\alpha)$ at positive height leaves only positive positions; setting $\nu = 1$ at height zero allows global sections. On $\mathbb{P}^n$, $h(-j) = j$ for $0 \leq j \leq n$, recovering classical $\mathbf{0}$ -regularity. On $\mathcal{H}_2$, Figure 4 and direct cohomology calculations give $\mathcal{R}_{\mathcal{H}_2}(\mathcal{O}) = \text{Nef}(\mathcal{H}_2) \cap \text{Pic}(\mathcal{H}_2)$.

Question 4.3. *For a smooth projective chamber X_i and a coherent sheaf $\mathcal{F}$, does the proposed definition satisfy*

- (1) nef persistence: $\mathcal{R}_i(\mathcal{F}) + \text{Nef}(X_i)_{\mathbb{Z}} \subseteq \mathcal{R}_i(\mathcal{F})$;
- (2) global generation: *if $\mathbf{d} \in \mathcal{R}_i(\mathcal{F})$, is $\mathcal{F}(\mathbf{d})$ globally generated?*

Nef translates of test line bundles can leave Z_X . I will seek resolutions using the original labels with controlled heights:

Conjecture 4.4. *Let X_i be a smooth projective chamber, and let G_i minimally generate its integral nef monoid. For every $\alpha \in \Theta_H$ and $L \in G_i$, there is an exact sequence*

$$0 \longrightarrow \mathcal{C}_N \longrightarrow \cdots \longrightarrow \mathcal{C}_0 \longrightarrow \mathcal{O}_{X_i}(L + \alpha) \longrightarrow 0,$$

where $\mathcal{C}_j$ is a finite sum of line bundles $\mathcal{O}_{X_i}(\beta)$ with $\beta \in \Theta_H$ and $\nu(\beta) \leq \nu(\alpha) + j$.

For $\mathbf{d} \in \mathcal{R}_i(\mathcal{F})$, tensoring by $\mathcal{F}(\mathbf{d})$ gives cohomology terms $H^{q+j}(\mathcal{F}(\mathbf{d} + \beta))$, which vanish for $q \geq \nu(\alpha)$ by the height bound; iteration would prove persistence. I will begin with toric Koszul

complexes on products and Hirzebruch surfaces, then study GKZ wall crossing. For global generation, I will analyze degree-zero terms in diagonal reconstruction and their evaluation maps.

Finally, Cranton Heller’s ceiling truncations and their HHL Fourier–Mukai description give complexes in the nef range that can retain higher module homology under arbitrarily ample twists [47]. I will study how this homology affects the comparison with MacLagan–Smith regularity, distinguishing homological behavior shared by the models from each model’s additional vanishing.

5. Extending Computational Algebra Software

Computation drives much of the recent progress in multigraded commutative algebra and toric geometry, and the PI has a track record of building such tools. The two projects below are modest in scope, but each produces a concrete tool for the community that also supports this proposal.

5.1 The toricStacks Package for Macaulay2 The lack of computational tools is a significant bottleneck in the study of homological invariants of stacks. A general framework is out of reach, but toric stacks are tractable: they correspond to *stacky fans*, whose well-developed combinatorics makes computation feasible [63, 64, 17], but is not implemented. Given the success of the *NormalToricVarieties* package, a *toricStacks* package should benefit the community.

Research Problem 5.1. *Develop a robust Macaulay2 package for computations with toric stacks.*

The package will (i) construct toric stacks from stacky fans and from families such as weighted projective stacks and root stacks, (ii) compute the Cox ring with its $\mathrm{Cl}(\mathcal{X})$ -grading and irrelevant ideal, and (iii) support coherent sheaves, including sheaf cohomology via local cohomology [56]. This is joint work in progress with Maya Banks and John Cobb; (i) is complete and (ii) is underway.

5.2 A Database of Human-Generated Examples in Algebra Good examples drive the development of conjectures, theorems, and algorithms, and a large trove of human-generated examples sits unused in the documentation and tests of open-source computer algebra systems. *Macaulay2* and its roughly 250 packages contain thousands of rings, ideals, modules, etc., nearly all chosen by experts to illustrate something interesting. Yet they are scattered across source files, not directly loadable, and impossible to sort or filter.

Research Problem 5.2. *Build a Macaulay2 package collecting the algebraic objects in its source and documentation into a queryable database with provenance and invariants.*

The package will parse the **EXAMPLE** and **TEST** blocks of installed packages, re-execute them in a sandbox, and record each resulting object as directly callable, with its source location and authorship. Much of the parsing already exists; the work lies in execution and curation. The result would be a testbed for conjectures and a benchmark suite easing future *Macaulay2* development. As calls grow for AI and machine learning tools built by mathematicians on mathematician-owned data, this is a small first step, with all authorship tracked and acknowledged.

6. Broader Impacts

6.1 Organizing. Building on Section 1.4.1, since 2022 I have organized *Gems of Combinatorics II* (30 participants), *AGNES* (150), and *BATMOBILE 2024 & 2025* (50/30), and I co-organize Dartmouth’s *Algebra & Number Theory* seminar. *Future Plans:* With co-organizers, I have proposed a *Thematic Semester in Combinatorial Algebraic Geometry* to the Fields Institute for 2030. I will host the next *Gems of Commutative Algebra* at Dartmouth in Fall 2028 and write a *Notices of the AMS* article on organizing early-career conferences. John Voight and I are in the early stages of planning a Macaulay2+Magma workshop on the future of mathematical software.

6.2 Open Educational Resources. In Fall 2026 I am lecturing for the Preliminary Arizona Winter School, a nine-week virtual course with 300 students (200 undergraduate and master's); all material (videos 100 pages of IBL notes and problems, etc) are freely available. Building on work of Karen Smith and Jack Jeffries, I designed an inquiry-based commutative algebra course of 16 worksheets (200+ pages). Both are on my website. *Future Plans:* I will develop these materials into an open-access introduction to algebraic geometry with Macaulay2 exercises.

6.3 Open Computational Resources. I also created *Syzygy Data* (syzygydata.com), a public database of large-scale syzygy computations, and co-authored four Macaulay2 packages. *Future Plans:* I have agreed to lead a project group aimed to further developing the `toricStacks` package at the 2027 *Macaulay2* conference being held at Auburn.

6.4 Campus Community. I am faculty advisor to Dartmouth's *AWM Student Chapter* and work with the campus *oSTEM* chapter to broaden participation in STEM. In Fall 2025 I founded *Unwritten Rules*, a graduate seminar that makes the hidden curriculum of an academic career explicit for students facing additional career challenges. *Future Plans:* I will continue running *Unwritten Rules* and find ways to include the postdocs at Dartmouth.

6.5 National & International Service. I have worked with the *AWM Executive Committee* to expand support for a broader range of mathematicians. From Winter 2023 until its dissolution in Spring 2025, I served on SLMath's *Committee on Women in Mathematics*. I now serve on the *AMS Committee on Engagement, Participation, & Advancement* and the *AMS Simons Travel Grant Committee*. *Future Plans:* In January 2026 I began a three-year term on the Board of Trustees of *Spectra*, which I previously served as inaugural president.

6.6 Mentoring. I advise three Ph.D. students on problems central to this proposal: Will D. (free resolutions and the moduli of spin curves), Jacob L-D. (regularity of torus-invariant subvarieties), and Alex M. (positivity of the Bondal–Thomsen collection). At MREG 2023 I began mentoring graduate students Bailee Z. (Michigan) and Jacob B. (KSU) on a project that led to [30]. In Summer 2026 I mentored three undergraduates through *Count Me In* (organized by M. Hering and D. Maclagan), and since Fall 2024 I have mentored two Dartmouth undergraduates. *Future Plans:* I will continue with *Count Me In* and run summer undergraduate projects built on computational experiments for Syzygy Data, with my Ph.D. students as near-peer mentors.

References

- [1] Dan Abramovich, Tom Graber, and Angelo Vistoli. Gromov–Witten theory of Deligne–Mumford stacks. *Amer. J. Math.*, 130(5):1337–1398, 2008.
- [2] Dan Abramovich and Angelo Vistoli. Compactifying the space of stable maps. *J. Amer. Math. Soc.*, 15(1):27–75, 2002.
- [3] Ayah Almousa, Juliette Bruce, Michael Loper, and Mahrud Sayrafi. The virtual resolutions package for Macaulay2. *J. Softw. Algebra Geom.*, 10(1):51–60, 2020.
- [4] Marian Aprodu, Gavril Farkas, Stefan Papadima, Claudiu Raicu, and Jerzy Weyman. Koszul modules and Green’s conjecture. *Invent. Math.*, 218(3):657–720, 2019.
- [5] Eran Assaf, Madeline Brandt, Juliette Bruce, Melody Chan, and Raluca Vlad. Tropicalizations of locally symmetric varieties, 2025. arXiv:2505.10504.
- [6] Matthew R. Ballard, Christine Berkesch, Michael K. Brown, Lauren Cranton Heller, Daniel Erman, David Favero, Sheel Ganatra, Andrew Hanlon, and Jesse Huang. King’s conjecture and the Cox category, 2024. arXiv:2501.00130.
- [7] Maya Banks, John Cobb, and Mahrud Sayrafi. Weighted syzygies of pointed curves, 2026. arXiv:2609.00312.
- [8] David Bayer and Michael Stillman. A criterion for detecting m -regularity. *Invent. Math.*, 87(1):1–11, 1987.
- [9] Dave Benson. Dickson invariants, regularity and computation in group cohomology. *Illinois Journal of Mathematics*, 48(1):171–197, 2004.
- [10] Christine Berkesch, Lauren Cranton Heller, Gregory G. Smith, and Jay Yang. Cellular free resolutions for normalizations of toric ideals, 2025. arXiv:2512.17871.
- [11] Christine Berkesch, Daniel Erman, and Gregory G. Smith. Virtual resolutions for a product of projective spaces. *Algebr. Geom.*, 7(4):460–481, 2020.
- [12] Aaron Bertram, Lawrence Ein, and Robert Lazarsfeld. Vanishing theorems, a theorem of Severi, and the equations defining projective varieties. *J. Amer. Math. Soc.*, 4(3):587–602, 1991.
- [13] Manjul Bhargava and Bjorn Poonen. The local-global principle for integral points on stacky curves. *J. Algebraic Geom.*, 31(4):773–782, 2022.
- [14] Christin Bibby, Melody Chan, Nir Gadish, and Claudia He Yun. Homology representations of compactified configurations on graphs applied to $\mathcal{M}_{2,n}$. *Exp. Math.*, 33(4):611–623, 2024.
- [15] Armand Borel. Stable real cohomology of arithmetic groups. *Ann. Sci. École Norm. Sup. (4)*, 7(2):235–272, 1974.
- [16] Lev Borisov and Zengrui Han. Hanlon–Hicks–Lazarev resolution revisited, 2025. arXiv:2509.11077.
- [17] Lev A. Borisov, Linda Chen, and Gregory G. Smith. The orbifold Chow ring of toric Deligne–Mumford stacks. *J. Amer. Math. Soc.*, 18(1):193–215, 2005.
- [18] Nicolás Botbol and Marc Chardin. Castelnuovo Mumford regularity with respect to multigraded ideals. *J. Algebra*, 474:361–392, 2017.
- [19] Daniel Bragg, Martin Olsson, and Rachel Webb. Ample vector bundles and moduli of tame stacks, 2024. arXiv:2407.01743.
- [20] Madeline Brandt, Juliette Bruce, Melody Chan, Margarida Melo, Gwyneth Moreland, and Corey Wolfe. On the top-weight rational cohomology of $\mathcal{A}_g$. *Geom. Topol.*, 28(2):497–538, 2024.
- [21] Madeline Brandt, Juliette Bruce, and Daniel Corey. The virtual Euler characteristic for binary matroids, 2023. arXiv:2301.10108.
- [22] Francis Brown. Bordifications of the moduli spaces of tropical curves and abelian varieties, and unstable cohomology of $\mathrm{GL}_g(\mathbb{Z})$ and $\mathrm{SL}_g(\mathbb{Z})$. *Invent. Math.*, 241(1):35–152, 2025.

- [23] Francis Brown, Melody Chan, Søren Galatius, and Sam Payne. Hopf algebras in the cohomology of $\mathcal{A}_g$, $\mathrm{GL}_n(\mathbb{Z})$, and $\mathrm{SL}_n(\mathbb{Z})$, 2024. arXiv:2405.11528.
- [24] Michael K. Brown and Daniel Erman. Linear strands of multigraded free resolutions. *Math. Ann.*, 390(2):2707–2725, 2024.
- [25] Michael K. Brown and Daniel Erman. Positivity and nonstandard graded Betti numbers. *Bull. Lond. Math. Soc.*, 56(1):111–123, 2024.
- [26] Michael K. Brown and Daniel Erman. A short proof of the Hanlon-Hicks-Lazarev theorem. *Forum Math. Sigma*, 12:e56, 2024.
- [27] Michael K. Brown and Daniel Erman. Linear syzygies of curves in weighted projective space. *Compos. Math.*, 161(4):916–944, 2025.
- [28] Michael K. Brown and Daniel Erman. Tate resolutions on toric varieties. *J. Eur. Math. Soc.*, 28(1):269–304, 2026.
- [29] Michael K. Brown and Mahrud Sayrafi. A short resolution of the diagonal for smooth projective toric varieties of Picard rank 2. *Algebra Number Theory*, 18(10):1923–1943, 2024.
- [30] Juliette Bruce, Jacob Bucciarelli, and Bailee Zacovic. Explorations of matroid complexes, 2026. arXiv:2605.24695.
- [31] Juliette Bruce, Daniel Corey, Daniel Erman, Steve Goldstein, Robert P. Laudone, and Jay Yang. Syzygies of $\mathbb{P}^1 \times \mathbb{P}^1$: data and conjectures. *J. Algebra*, 593:589–621, 2022.
- [32] Juliette Bruce, Lauren Cranton Heller, and Mahrud Sayrafi. Bounds on multigraded regularity, 2022. arXiv:2208.11115.
- [33] Juliette Bruce, Lauren Cranton Heller, and Mahrud Sayrafi. Characterizing multigraded regularity and virtual resolutions on products of projective spaces. *Advances in Mathematics*, 500:111049, 2026.
- [34] Juliette Bruce, Lauren Cranton Heller, Mahrud Sayrafi, and Alexandra Seceleanu. Seshadri regions and the asymptotic shape of multigraded regularity, 2025. arXiv:2512.05289.
- [35] Juliette Bruce, Daniel Erman, Steve Goldstein, and Jay Yang. Conjectures and computations about Veronese syzygies. *Exp. Math.*, 29(4):398–413, 2020.
- [36] Charles Cadman. Using stacks to impose tangency conditions on curves. *Amer. J. Math.*, 129(2):405–427, 2007.
- [37] Samir Canning, Hannah Larson, and Sam Payne. The eleventh cohomology group of $\overline{\mathcal{M}}_{g,n}$. *Forum Math. Sigma*, 11:Paper No. e62, 18 pp., 2023.
- [38] Guido Castelnuovo. Sui multipli di una serie lineare di gruppi di punti appartenente ad una curva algebrica. *Rend. Circ. Mat. Palermo*, 7:89–110, 1893.
- [39] Giulio Caviglia, Marc Chardin, Jason McCullough, Irena Peeva, and Matteo Varbaro. Regularity of prime ideals. *Math. Z.*, 291(1-2):421–435, 2019.
- [40] Melody Chan, Søren Galatius, and Sam Payne. Tropical curves, graph complexes, and top weight cohomology of $\mathcal{M}_g$. *J. Amer. Math. Soc.*, 34(2):565–594, 2021.
- [41] Melody Chan, Søren Galatius, and Sam Payne. Topology of moduli spaces of tropical curves with marked points. In Paolo Aluffi, David Anderson, Milena Hering, Mircea Mustață, and Sam Payne, editors, *Facets of Algebraic Geometry: A Collection in Honor of William Fulton’s 80th Birthday. Vol. 1*, volume 472 of *London Mathematical Society Lecture Note Series*, pages 77–131. Cambridge University Press, 2022.
- [42] Marc Chardin and Navid Nemati. Multigraded regularity of complete intersections, 2020. arXiv:2012.14899.
- [43] Alessandro Chiodo, David Eisenbud, Gavril Farkas, and Frank-Olaf Schreyer. Syzygies of torsion bundles and the geometry of the level ℓ modular variety over $\overline{\mathcal{M}}_g$. *Invent. Math.*, 194(1):73–118, 2013.
- [44] John Cobb. Syzygies of curves in products of projective spaces. *Math. Z.*, 306(2):Paper No. 27, 19 pp., 2024.

- [45] Kaelin Cook-Powell and David Jensen. Tropical methods in Hurwitz-Brill-Noether theory. *Adv. Math.*, 398:Paper No. 108199, 42, 2022.
- [46] David A. Cox. The homogeneous coordinate ring of a toric variety. *J. Algebraic Geom.*, 4(1):17–50, 1995.
- [47] Lauren Cranton Heller. Explicit constructions of short virtual resolutions of truncations. *Bulletin of the London Mathematical Society*, 57(12):4000–4014, 2025.
- [48] S. Dale Cutkosky, Jürgen Herzog, and Ngô Việt Trung. Asymptotic behaviour of the Castelnuovo-Mumford regularity. *Compositio Math.*, 118(3):243–261, 1999.
- [49] Steven Dale Cutkosky, Lawrence Ein, and Robert Lazarsfeld. Positivity and complexity of ideal sheaves. *Math. Ann.*, 321(2):213–234, 2001.
- [50] Caitlin M. Davis and Boyana Martinova. The Koszul property for truncations of nonstandard graded polynomial rings, 2025. arXiv:2503.17541.
- [51] Michel Demazure. Anneaux gradués normaux. In *Introduction à la théorie des singularités, II*, volume 37 of *Travaux en Cours*, pages 35–68. Hermann, Paris, 1988.
- [52] Anand Deopurkar. The canonical syzygy conjecture for ribbons. *Math. Z.*, 288(3-4):1157–1164, 2018.
- [53] Lawrence Ein and Robert Lazarsfeld. Syzygies and Koszul cohomology of smooth projective varieties of arbitrary dimension. *Invent. Math.*, 111(1):51–67, 1993.
- [54] David Eisenbud, Daniel Erman, and Frank-Olaf Schreyer. Tate resolutions for products of projective spaces. *Acta Math. Vietnam.*, 40(1):5–36, 2015.
- [55] David Eisenbud and Shiro Goto. Linear free resolutions and minimal multiplicity. *J. Algebra*, 88(1):89–133, 1984.
- [56] David Eisenbud, Mircea Mustață, and Mike Stillman. Cohomology on toric varieties and local cohomology with monomial supports. *J. Symbolic Comput.*, 29(4–5):583–600, 2000.
- [57] Gavril Farkas and Michael Kemeny. The generic Green-Lazarsfeld secant conjecture. *Invent. Math.*, 203(1):265–301, 2016.
- [58] Gavril Farkas and Michael Kemeny. The Prym-Green conjecture for torsion line bundles of high order. *Duke Math. J.*, 166(6):1103–1124, 2017.
- [59] Gavril Farkas, Mircea Mustață, and Mihnea Popa. Divisors on $\mathcal{M}_{g,g+1}$ and the minimal resolution conjecture for points on canonical curves. *Ann. Sci. École Norm. Sup. (4)*, 36(4):553–581, 2003.
- [60] David Favero and Jesse Huang. Rouquier dimension is Krull dimension for normal toric varieties. *Eur. J. Math.*, 9(4):Paper No. 91, 2023.
- [61] David Favero and Jesse Huang. Homotopy path algebras. *Selecta Mathematica, New Series*, 31(2):25, 2025.
- [62] David Favero and Mykola Sapronov. Line bundle resolutions via the coherent-constructible correspondence, 2024. arXiv:2411.17873.
- [63] Anton Geraschenko and Matthew Satriano. Toric stacks I: The theory of stacky fans. *Trans. Amer. Math. Soc.*, 367(2):1033–1071, 2015.
- [64] Anton Geraschenko and Matthew Satriano. Toric stacks II: Intrinsic characterization of toric stacks. *Trans. Amer. Math. Soc.*, 367(2):1073–1094, 2015.
- [65] Gerd Gotzmann. Eine bedingung für die flachheit und das Hilbertpolynom eines gradierten Ringes. *Math. Z.*, 158(1):61–70, 1978.
- [66] Mark L. Green. Koszul cohomology and the geometry of projective varieties. *J. Differential Geom.*, 19(1):125–171, 1984. With an appendix by Mark L. Green and Robert Lazarsfeld.
- [67] Alexander Grothendieck. Techniques de construction et théorèmes d’existence en géométrie algébrique IV: les schémas de Hilbert. In *Séminaire Bourbaki: années 1960/61, exposés 205–222*, number 6 in Séminaire Bourbaki, pages 249–276. Société Mathématique de France, 1961. Exposé 221.

- [68] Samuel Grushevsky. Geometry of $\mathcal{A}_g$ and its compactifications. In *Algebraic geometry—Seattle 2005. Part 1*, volume 80, Part 1 of *Proc. Sympos. Pure Math.*, pages 193–234. Amer. Math. Soc., Providence, RI, 2009.
- [69] Laurent Gruson, Robert Lazarsfeld, and Christian Peskine. On a theorem of Castelnuovo, and the equations defining space curves. *Invent. Math.*, 72(3):491–506, 1983.
- [70] Richard Hain. The rational cohomology ring of the moduli space of abelian 3-folds. *Math. Res. Lett.*, 9(4):473–491, 2002.
- [71] Zengrui Han. Minimal resolutions of toric substacks by line bundles, 2026. arXiv:2604.14384.
- [72] Andrew Hanlon, Jeff Hicks, and Oleg Lazarev. Resolutions of toric subvarieties by line bundles and applications. *Forum Math. Pi*, 12:Paper No. e24, 58, 2024.
- [73] J. William Hoffman and Hao Hao Wang. Castelnuovo–Mumford regularity in biprojective spaces. *Adv. Geom.*, 4(4):513–536, 2004.
- [74] Klaus Hulek and Orsola Tommasi. Cohomology of the second Voronoi compactification of $\mathcal{A}_4$. *Doc. Math.*, 17:195–244, 2012.
- [75] Klaus Hulek and Orsola Tommasi. The topology of $\mathcal{A}_g$ and its compactifications. In Jan Arthur Christoffersen and Kristian Ranestad, editors, *Geometry of moduli*, volume 14 of *Abel Symp.*, pages 135–193. Springer, Cham, 2018. With an appendix by Olivier Taïbi.
- [76] Jun-ichi Igusa. On Siegel modular forms of genus two. *Amer. J. Math.*, 84(1):175–200, 1962.
- [77] David Jensen and Dhruv Ranganathan. Brill-Noether theory for curves of a fixed gonality. *Forum Math. Pi*, 9:Paper No. e1, 33, 2021.
- [78] Michael Kemeny. Universal secant bundles and syzygies of canonical curves. *Invent. Math.*, 223(3):995–1026, 2021.
- [79] Michael Kemeny. The rank of syzygies of canonical curves. *J. Reine Angew. Math.*, 810:97–138, 2024.
- [80] Andrew Kobin and David Zureick-Brown. Wild stacky curves and rings of mod p modular forms, 2025. arXiv:2510.08821.
- [81] Vijay Kodiyalam. Asymptotic behaviour of Castelnuovo–Mumford regularity. *Proc. Amer. Math. Soc.*, 128(2):407–411, 2000.
- [82] Sijong Kwak. Castelnuovo regularity for smooth subvarieties of dimensions 3 and 4. *J. Algebraic Geom.*, 7(1):195–206, 1998.
- [83] Sijong Kwak. Generic projections, the equations defining projective varieties and Castelnuovo regularity. *Math. Z.*, 234(3):413–434, 2000.
- [84] Eric Larson, Hannah Larson, and Isabel Vogt. Global Brill–Noether theory over the Hurwitz space. *Geom. Topol.*, 29(1):193–257, 2025.
- [85] Hannah K. Larson. Refined Brill–Noether theory for all trigonal curves. *Eur. J. Math.*, 7(4):1524–1536, 2021.
- [86] Robert Lazarsfeld. A sharp Castelnuovo bound for smooth surfaces. *Duke Math. J.*, 55(2):423–429, 1987.
- [87] Diane Maclagan and Gregory G. Smith. Multigraded Castelnuovo–Mumford regularity. *J. Reine Angew. Math.*, 571:179–212, 2004.
- [88] Diane Maclagan and Gregory G. Smith. Uniform bounds on multigraded regularity. *J. Algebraic Geom.*, 14(1):137–164, 2005.
- [89] Boyana Martinova. Asymptotic syzygies of weighted projective spaces, 2025. arXiv:2510.12708.
- [90] Jason McCullough and Irena Peeva. Counterexamples to the Eisenbud–Goto regularity conjecture. *J. Amer. Math. Soc.*, 31(2):473–496, 2018.
- [91] Jason McCullough and Irena Peeva. The regularity conjecture for prime ideals in polynomial rings. *EMS Surv. Math. Sci.*, 7(1):173–206, 2020.
- [92] David Mumford. *Geometric invariant theory*, volume 34 of *Ergebnisse der Mathematik und ihrer Grenzgebiete, (N.F.)*. Springer-Verlag, Berlin-New York, 1965.

- [93] Brett Nasserden and Stanley Yao Xiao. The density of rational points on $\mathbb{P}^1$ with three stacky points, 2020. arXiv:2011.06586.
- [94] Sam Payne and Thomas Willwacher. Weight two compactly supported cohomology of moduli spaces of curves. *Duke Math. J.*, 173(16):3107–3178, 2024.
- [95] Nathan Pflueger. Brill-Noether varieties of k -gonal curves. *Adv. Math.*, 312:46–63, 2017.
- [96] Henry C. Pinkham. Normal surface singularities with $\mathbb{C}^*$ action. *Math. Ann.*, 227(2):183–193, 1977.
- [97] Henry C. Pinkham. A Castelnuovo bound for smooth surfaces. *Invent. Math.*, 83(2):321–332, 1986.
- [98] Jessica Sidman, Adam Van Tuyl, and Haohao Wang. Multigraded regularity: coarsenings and resolutions. *Journal of Algebra*, 301(2):703–727, 2006.
- [99] Irena Swanson. Powers of ideals: primary decompositions, Artin–Rees lemma and regularity. *Math. Ann.*, 307(2):299–313, 1997.
- [100] Peter Symonds. On the Castelnuovo–Mumford regularity of the cohomology ring of a group. *Journal of the American Mathematical Society*, 23(4):1159–1173, 2010.
- [101] Peter Symonds. On the Castelnuovo–Mumford regularity of rings of polynomial invariants. *Annals of Mathematics*, 174(1):499–517, 2011.
- [102] John Voight and David Zureick-Brown. The canonical ring of a stacky curve. *Mem. Amer. Math. Soc.*, 277(1362):v+144, 2022.
- [103] Claire Voisin. Green’s generic syzygy conjecture for curves of even genus lying on a $K3$ surface. *J. Eur. Math. Soc. (JEMS)*, 4(4):363–404, 2002.
- [104] Claire Voisin. Green’s canonical syzygy conjecture for generic curves of odd genus. *Compos. Math.*, 141(5):1163–1190, 2005.
- [105] Keiichi Watanabe. Some remarks concerning Demazure’s construction of normal graded rings. *Nagoya Math. J.*, 83:203–211, 1981.